

The Lord of the Ring?

Rings with sensors built into them are not new, at least conceptually. There are quite a handful of start-ups that have tried to build these in one way or another.

Business Week reports that CES 2015 saw a big crowd at the display for Logbar's product called Ring. Previously seen on crowdfunding website, *Kickstarter*, where it raised US\$ 880,000, these now have enough *moolah* to display at CES.

But, is the product ready yet? Their initial try was a failure with rings that were deemed not worthy of funders' fingers due to the material used (zinc) and functionality issues. Logbar says, it is ready now and will be out in March 2015, but Joshua Bernstein reported in his article from CES 2015 about spotting a disclaimer warning of potential itchiness, irritation and rashes, depending on the user's health conditions.

On the other side, there is our own *desi* team building Fin, a ring that can be used as a gesture interface. While it was successfully campaigned on crowdfunding website, *Indiegogo*, last year with approximately US\$ 202,000 raised, the company has missed its target delivery date of September 2014, as the product remains under development. The team has posted an updated delivery date of May 2015 in response to queries from funders about the product's status. They have also uploaded a video of the prototype in its current stage, which is reassuring. However, it still remains to be seen how the final product will hit consumer expectations.

Did you know there are people who can sense their laptops begin to stall, or if their hard drives have begun to spin-up under load?

small electronics pack clips on to the garment, where data can be processed and made sense of.

According to a paper published in *Advanced Function Materials*, researchers at Singhua University and Nanchang University in China have claimed that it is possible to use grapheme-strain sensors to monitor vital signs on a human body. This technology, once developed, could be implemented into implantable devices or on electronics skin. For instance, the strength of motion created when a person moves creates a larger strain on the sensor, which allows the signal to be recorded and monitored easily.

Using wearables as the first information tool in sports is another trend that is picking up, and about time too. Last year's FIFA World Cup Final saw one example, where a German player was injured after getting a shoulder-hit to the head. The player played for some time after the hit, before it was realised that he had a serious injury.

Blackbox Biometrics and MC10 are two companies that have been

building technology for this sector. By measuring the magnitude of motion and rotational forces in the environment around the brain, these sensors can give an idea about the actual impact to the brain and the chances of potential injury.

University of Michigan is working on millimetre-scale energy-harvesting sensors, such as a wireless intra-ocular pressure sensor that is intended for glaucoma patients to measure and report the pressure in the eye. "All elements, including the sensor, solar cell, energy storage, MCU and radio, could be part of this sensor. The university has indicated that it occupies a volume of only 1.5mm³ (0.5x1.5x2.0mm³) and can be implanted with a minimally-invasive procedure that is routinely used for cataract surgery. Commercialisation is anticipated in two to three years," adds T. Anand, managing director, Knewron.

Smarter algorithms. Sensing data alone is just halfway to the goal when it comes to wearable devices, because the amount of noise in an everyday-use environment is pretty high. PerformTek sensor module technology, mentioned earlier in this story, has a signal-extraction method that removes noise from skin movement and footsteps, as well as the environment, to keep data clean. Additional algorithms are then run

on this to accurately extract heart rate, respiration rate and other parameters. An additional proprietary algorithm leverages accelerometer data to measure running and cycling cadence, calories burned and distance travelled.

But, why did it take so long for these applications to be tapped?

"Electronics are rigid and boxy, while humans are soft and curvy."

First heard at the D11 conference in 2013, this statement is a simple way to understand the challenge put forward by wearables. You need circuits that can follow the form and elasticity of its user, yet keep functioning optimally.

MC10 and Multek are examples of companies that seem to have understood the limitation of conventional electronic board technology when considering their utility for wearable-device designs. One of MC10's patents is for what is termed as extremely stretchable electronics. The invention is built around the idea of using two isolated electronic components attached to an elastic-substrate material. An electrical interconnection is placed between components that maintain electrical characteristics, while the substrate is stretched.

An interesting MC10 design is Biostamp, a sensing sticker technology that can be placed on the skin. It stretches and flexes along with the skin, while maintaining the functionality designed into it. It can measure a variety of physiological functions, like data from the brain, muscles, heart, body temperature and movement.

Flexible printed circuit board (PCB) is a technology that has been around for some time. Multek has expertise in manufacturing flexible PCBs, and available technologies include standard flex and rigid flex, and also newer technology, like Novaflex HDI that features high-density interconnects. Novaclad is a material from Sheldahl, which is made-up of ultra-thin build-up of copper materials for flexible circuits.

Newer technology designed especially for wearables is the fabric circuit board. Developed by Xiao-Ming Tao and her team from Institute of Textiles